

Wenzhi Fang

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RESEARCH INTERESTS

My research broadly focuses on **large language models (LLMs)**, spanning **reinforcement learning (RL)-based post-training and reasoning**, **LLM agents and multi-agent collaboration**, and **efficient fine-tuning and deployment** in distributed and on-device settings. I am also interested in **optimization**, especially **distributed optimization** for machine learning.

EDUCATION

- **Purdue University** Aug. 2023 – May 2027 (expected)
Ph.D. Candidate in Electrical and Computer Engineering
◦ **Advisor:** Prof. [Christopher G. Brinton](#) West Lafayette, IN, USA
- **ShanghaiTech University** Sept. 2020 – Jul. 2023
M.S. in Communication and Information Systems
◦ **Advisors:** Prof. [Yong Zhou](#) and Prof. [Yuanming Shi](#) Shanghai, China
- **Shanghai University** Sept. 2016 – Jul. 2020
B.S. in Communication Engineering Shanghai, China

INTERNSHIP

- **Samsung Research America (SRA)** May 2026 – Aug. 2026
Research Intern @ AI Center (Mentor: Dr. [Nicholas Tzou](#)) Mountain View, CA, USA
Developed Null-Basis LoRA (NB-LoRA) to adapt RL-trained LLMs while preserving reasoning capabilities by constraining updates to approximate null spaces of reasoning activations. Matched standard LoRA's downstream adaptation performance while retaining reasoning accuracy near pre-fine-tuning levels, including on held-out reasoning benchmarks.
💠 [LLM Reasoning](#) 💠 [Parameter-Efficient Fine-Tuning](#) 💠 [Continual Learning](#)
- **King Abdullah University of Science and Technology (KAUST)** Aug. 2022 – Feb. 2023
Research Intern @ Machine Learning and Optimization Lab (Advisor: Prof. [Peter Richtárik](#)) Thuwal, Saudi Arabia
Studied a primal-dual hybrid gradient descent algorithm and explored variance-reduction techniques for primal-dual optimization.
💠 [Primal-Dual Optimization](#) 💠 [Variance Reduction](#)

SELECTED PUBLICATIONS

- [NeurIPS 2026] **Iterative Critique-and-Routing Controller for Multi-Agent Systems with Heterogeneous LLMs**
W. Fang, L. Yuan, D.-J. Han, G. Lan, and C. G. Brinton
[PDF](#) | We propose a multi-turn critique-and-routing controller that improves coordination among heterogeneous LLM agents by iteratively routing and refining responses while balancing agent utilization.
💠 [Multi-Agent Systems](#) 💠 [Multi-Turn Interaction](#) 💠 [Reinforcement Learning](#)
- [NeurIPS 2026] **PAAC: Privacy-Aware Agentic Device-Cloud Collaboration**
L. Yuan, **W. Fang**, S. Wang, and C. G. Brinton [PDF](#)
💠 [LLM Agents](#) 💠 [Privacy](#)
- [ICML 2026] **Bridging On-Device and Cloud LLMs for Collaborative Reasoning**
W. Fang, D.-J. Han, L. Yuan, and C. G. Brinton
[PDF](#) | We propose a unified reinforcement learning framework that empowers on-device LLMs to autonomously learn both problem-solving reasoning and routing policies, deciding when to process queries locally or to offload them to cloud LLMs, removing the need for separate routing classifiers.
💠 [LLM Reasoning](#) 💠 [Reinforcement Learning](#) 💠 [Device-Cloud Collaboration](#)
- [ICML 2026] **Federated Sketching LoRA: On-Device Collaborative Fine-Tuning of Large Language Models**
W. Fang, D.-J. Han, L. Yuan, S. Hosseinalipour, and C. G. Brinton
[PDF](#) | We propose sketching LoRA, a theoretically grounded parameter-efficient fine-tuning methodology that retains LoRA's

flexibility while adapting to the communication and computational abilities of individual devices.

◆ [Parameter-Efficient Fine-Tuning](#) ◆ [Federated Learning](#)

[NeurIPS 2024] **Hierarchical Federated Learning with Multi-Timescale Gradient Correction**

W. Fang, D.-J. Han, E. Chen, S. Wang, and C. G. Brinton

[PDF](#) | We propose an algorithm addressing multi-level data heterogeneity in hierarchical federated learning (fog learning), deriving strong theoretical results without additional data-heterogeneity assumptions and filling a critical gap in the federated learning literature.

◆ [Federated Learning](#) ◆ [Distributed Optimization](#)

[IEEE/ACM ToN 2025] **Federated Learning over Hierarchical Wireless Networks: Training Latency Minimization via Submodel Partitioning**

W. Fang, D.-J. Han, and C. G. Brinton

[PDF](#) | We propose hierarchical independent submodel training (HIST), a federated learning methodology that caters to resource-constrained mobile devices in hierarchical wireless networks while minimizing training latency.

◆ [Federated Learning](#) ◆ [Wireless Networks](#)

[IEEE TSP 2022] **Communication-Efficient Stochastic Zeroth-Order Optimization for Federated Learning**

★ **Highly Cited Paper**

W. Fang, Z. Yu, Y. Jiang, Y. Shi, C. Jones, and Y. Zhou

[PDF](#) | We propose FedZO, a federated zeroth-order algorithm with convergence guarantees that makes training forward-only and eliminates the memory overhead of backpropagation; it has been widely applied in LLM research, e.g., LLM attack, prompt tuning, and efficient fine-tuning.

◆ [Zeroth-Order Optimization](#) ◆ [Federated Learning](#)

PREPRINTS

[Preprint 2026] **Reasoning-Preserving Fine-Tuning of Post-RL LLMs with Null-Basis LoRA**

W. Fang, N. Tzou, L. Valkov, and S. Chappidi

[PDF](#) | We propose Null-Basis LoRA (NB-LoRA), which constrains fine-tuning updates to approximate null spaces of reasoning activations to preserve reasoning in post-RL LLMs, matching standard LoRA's downstream adaptation performance while maintaining reasoning accuracy near pre-fine-tuning levels.

◆ [LLM Reasoning](#) ◆ [Parameter-Efficient Fine-Tuning](#) ◆ [Continual Learning](#)

[Preprint 2026] **Joint Continual Learning of Local Language Models and Cloud Offloading Decisions with Budget Constraints**

E. Chen*, W. Fang*, S. Wang, L. Yuan, and C. G. Brinton (*equal contribution)

[PDF](#) | We propose a constraint-aware RL method that embeds cloud-usage regulation into advantage computation, enabling stable, budget-controlled local-cloud LLM collaboration and reduced catastrophic forgetting under continual learning without per-task reward tuning.

◆ [Continual Learning](#) ◆ [Reinforcement Learning](#) ◆ [Device-Cloud Collaboration](#)

[Preprint 2026] **Self-Play Enhancement via Advantage-Weighted Refinement in Online Federated LLM Fine-Tuning with Real-Time Feedback**

J. Lee, W. Fang, D.-J. Han, S. Hosseinalipour, and C. G. Brinton [PDF](#)

◆ [Federated Learning](#) ◆ [LLM Fine-Tuning](#)

ACADEMIC SERVICES

Reviewer — Conferences: NeurIPS, ICML, ICLR, AISTATS, AAAI

Reviewer — Journals / Transactions: IEEE/ACM Transactions on Networking (ToN); IEEE Transactions on Signal Processing (TSP); IEEE Transactions on Mobile Computing (TMC); Transactions on Machine Learning Research (TMLR)

TECHNICAL SKILLS

Languages: Python, C, C++, MATLAB

Tools: PyTorch, TensorFlow, vLLM, VERL, Git, $\LaTeX$